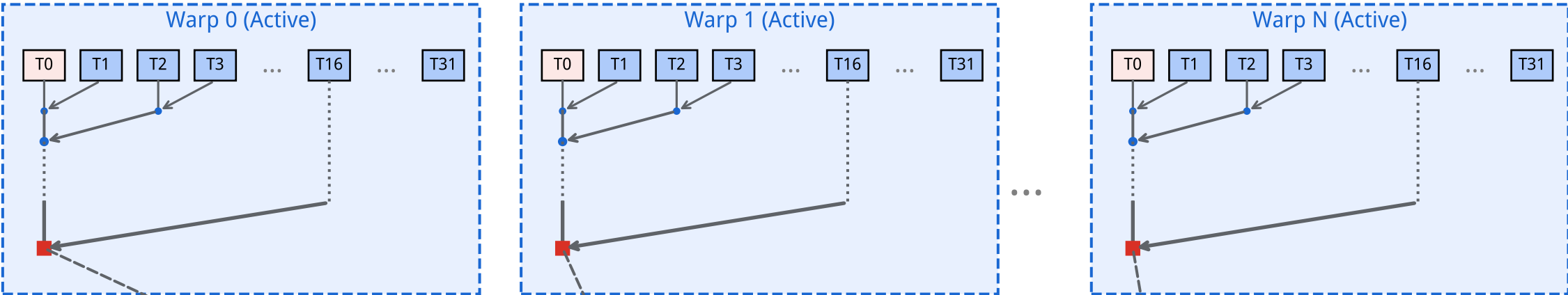


第一阶段：Warp级内部无锁蝶形规约



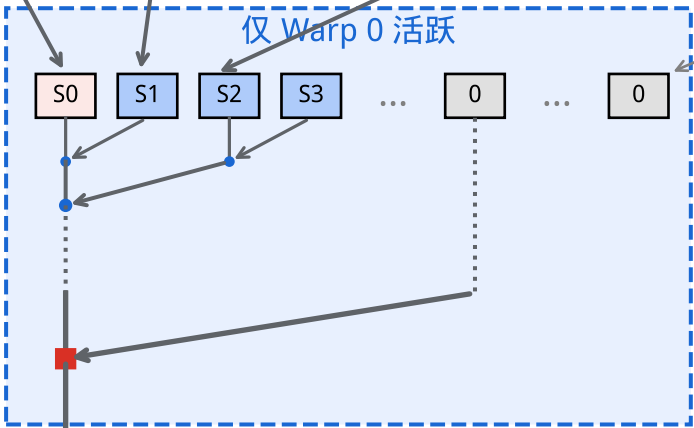
第二阶段：轻量级Shared Memory暂存与全局同步

Leader线程(Lane 0)写入落盘



强制全局同步屏障: `__syncthreads()`

第三阶段：单一Warp唤醒规约



越界元素进行
Zero-padding

